12b b) Implement a chat program using sockets programming

Aim:
To Implement a chat program using code sockets program

procedure
1) To Implement the UDP code.

```
Import socket
Import threading
#... Server ...
def udp_Server():
  S = Socket.Socket(Socket.AF_INET, SOCKET.SOCK_DR
  S.bind(('127.0.0.1', 12000))
  print("[Server] Running on 127.0.0.1:12000")
  while True:
    data, addr = S.recvfrom(1024)
    msg = data.decode()
    if msg.lower() == 'quit':
print(f"[Server] Client {addr} ended the
Connect.").
    break
print(f"[Server] Recieved from {addr}: {msg}
S.sendto(data, addr)
S.close
print("[Server] Stopped.")
```

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```
# client :
def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[client] closing connection...")
            break
        data, _ = c.recvfrom(1024)
        print("Echo from Server", data.decode())
    c.close()
    print("[client] Stopped.")

# --- Main ---
if __name__ == "__main__":
    threading.Thread(target=udp_server,
    daemon=True).start()
    udp_client()
```

Output:

Enter message (or 'quit' to stop): [Server]

Running on 127.0.0.1:12000hi

[Server] Received from ('127.0.0.1', 49670): hi

Echo from Server: hi

Enter message (or 'quit' to stop): ('127.0.0.1', 49670):

[client] closing connection -- [Server] client ('127.0.0.1', 49670) ended the connection.

[client] stopped [server] stopped.

Result:

Therefore, to implement a chat program using Socket Programming executed successfully.

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